

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Expands the bracket and obtains any correct form (PI)	AO1.1b	B1	$f'(x) = 4x^2 - 12 + \frac{9}{x^2}$ $f(x) = \int 4x^2 - 12 + \frac{9}{x^2} dx$ $= \frac{4x^3}{3} - 12x - \frac{9}{x} + c$ $f(3) = 36 - 36 - 3 + c = 2$ $c = 5$ $f(x) = \frac{4x^3}{3} - 12x - \frac{9}{x} + 5$
Integrates at least one of 'their' terms correctly	AO1.1a	M1	
Substitutes $x = 3$ into 'their' integrated expression and equates to 2	AO1.1a	M1	
Obtains completely correct expression, must be explicitly stated CAO (ACF)	AO1.1b	A1	
			Total 4 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Begins a proof using a valid method Eg. Factor theorem, algebraic division, multiplication of correct factors	AO1.1a	M1	$p\left(-\frac{1}{2}\right) = 30 \times \left(-\frac{1}{2}\right)^3 - 7\left(-\frac{1}{2}\right)^2 - 7\left(-\frac{1}{2}\right) + 2$ $= 0$ $\therefore 2x + 1 \text{ is a factor of } p(x)$
	Constructs rigorous mathematical proof. To achieve this mark: Factor theorem the student must clearly substitute and state that $p(-1/2) = 0$ and clearly state that this implies that $2x + 1$ is a factor Algebraic division OR Multiplication of correct factors The method must be completely correct with a concluding statement	AO2.1	R1	
(b)	Obtains quadratic factor PI	AO1.1a	M1	$p(x) = (2x + 1)(15x^2 - 11x + 2)$

(c)	Obtains second linear factor	AO1.1b	A1	$= (2x + 1)(5x - 2)(3x - 1)$
	Writes $p(x)$ as the product of the correct three linear factors. NMS correct answer 3/3	AO1.1b	A1	
	Rearranges to achieve a cubic equation in $\sec x$ (or $\cos x$)	AO3.1a	M1	$\frac{30\sec^2 x + 2\cos x}{7} = \sec x + 1$
	Equates to zero and uses result from (b) or factorises	AO1.1a	M1	$\Rightarrow 30 \sec^2 x + 2 \cos x = 7 \sec x + 7$
	Deduces that if solutions exist they must be of the form $\sec x = -\frac{1}{2}$, $\sec x = \frac{1}{3}$ or $\sec x = \frac{2}{5}$ OE	AO2.2a	A1	$\Rightarrow 30 \sec^3 x + 2 = 7 \sec^2 x + 7 \sec x$ $30 \sec^3 x - 7 \sec^2 x - 7 \sec x + 2 = 0$
	Explains that the range of $\sec x$ is $(-\infty, -1] \cup [1, \infty)$ OE OE for $\cos x$	AO2.4	E1	$\Rightarrow (2 \sec x + 1)(5 \sec x - 2)(3 \sec x - 1) = 0$ $\Rightarrow \sec x = -\frac{1}{2}, \frac{1}{3}, \frac{2}{5}$
	Completes argument explaining that there cannot be any real solutions as values are outside of the function's range	AO2.1	R1	These values do not fall within the range of $\sec x$ as they are between -1 and 1 $\therefore \frac{30\sec^2 x + 2\cos x}{7} = \sec x + 1$ has no real solutions.
				Total 10 marks

Q3.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	9
Total 1 mark			

Q4.

(a)	Marking Instructions	AO	Marks	Typical Solution
	Demonstrates $p(-3) = 0$	AO1.1b	B1	$p(-3) = 2(-3)^3 + 7(-3)^2 + 2(-3) - 3$ $= -54 + 63 - 6 - 3 = 0$ $p(-3) = 0 \Rightarrow x + 3$ is a factor
	Constructs rigorous mathematical proof (to achieve this mark, the student must clearly calculate and state that	AO2.1	R1	

	p(-3) = 0 and clearly state that this implies that $x + 3$ is a factor)			
(b)	Factorises the numerator and denominator (this mark is achieved for any reasonable attempt at factorisation through the selection of an appropriate method, eg long division)	AO1.1a	M1	$\frac{(x+3)(2x^2+x-1)}{(2x+1)(2x-1)}$ $= \frac{(x+3)(2x-1)(x+1)}{(2x+1)(2x-1)}$ $= \frac{(x+3)(x+1)}{(2x+1)}, x \neq \pm \frac{1}{2}$
	Finds second factor in numerator or fully factorises denominator (PI by complete factorisation)	AO1.1b	A1	
	Finds fully correct factorised expression (PI by complete factorisation)	AO1.1b	A1	
	Obtains a completely correct solution with restriction on domain stated	AO1.1b	A1	
				Total 6 marks

Q5.

- (a) (i) $p(3) = 3^3 - 4 \times 3^2 - 3 \times 3 + 18$
*p(3) attempted **not** long division*

M1

$p(3) = 27 - 36 - 9 + 18 = 0 \Rightarrow x - 3$ is a factor
shown = 0 plus statement

A1

2

- (ii) Quadratic factor $(x^2 - x + c)$ or $(x^2 + bx - 6)$
-x or -6 term by inspection
or full long division by $x - 3$
or comparing coefficients
or p(-2) attempted

M1

Quadratic factor $(x^2 - x - 6)$
correct quadratic factor (or $x + 2$ shown to be factor by Factor Theorem)

A1

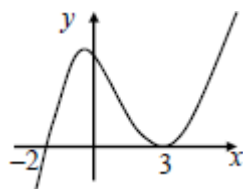
$$[p(x)=] (x-3)(x-3)(x+2)$$

or $[p(x)=] (x-3)^2(x+2)$
must see product of factors

A1

3

(b)



cubic curve with one maximum and one minimum

M1

meeting x-axis at -2 and touching x-axis at 3

A1

Final A1 is dependent on previous A1 and can be withheld if curve has very poor curvature beyond $x = 3$, V shape at $x = 3$ etc

graph as shown, going beyond $x = -2$ but condone max on or to right of y-axis

A1

3

[8]

Q6.

(a) $f\left(-\frac{1}{2}\right) = 2\left(-\frac{1}{2}\right)^3 + \left(-\frac{1}{2}\right)^2 - 8\left(-\frac{1}{2}\right) - 7$

Evaluate $f\left(-\frac{1}{2}\right)$, not long division.

M1

$$= -3$$

A1

2

(b) (i) $g\left(-\frac{1}{2}\right) = 0 \Rightarrow -3 + d = 0$

Or $f\left(-\frac{1}{2}\right) + d = 0$

$$d = 3 \Rightarrow g(x) = 2x^3 + 2x^2 - 8x - 7 + 3$$

$$g(x) = 2x^3 + 2x^2 - 8x - 4$$

All steps seen with conclusion

AG

B1

*Allow verification with $-\frac{1}{4} + \frac{1}{4} + 4 - 4 = 0$ seen,
and conclusion; therefore factor*

1

$$(ii) \quad g(x) = 2x^3 + 2x^2 - 8x - 4 = (2x + 1)(x^2 - 4) = (2x + 1)(x + 2)(x - 2)$$

$$a = -4$$

B1

1

$$(iii) \quad 2x^3 - 3x^2 - 2x = x(2x + 1)(x - 2)$$

Clear attempt to factorise denominator; 3 factors needed.

M1

$$\frac{(2x+1)(x+2)(x-2)}{x(2x+1)(x-2)} = \frac{x+2}{x}$$

At least one correct factor cancelled

m1

$$\frac{g(x)}{2x^3 - 3x^2 - 2x} = 1 + \frac{2}{x}$$

CSO part (a)(iii) NMS is 0 / 3

A1

3

Alternative:

$$\frac{g(x)}{2x^3 - 3x^2 - 2x} = 1 + \frac{4x^2 - 6x - 4}{2x^3 - 3x^2 - 2x}$$

$$1 + \frac{\text{quadratic}}{2x^3 - 3x^2 - 2x}$$

(M1)

$$= 1 + \frac{2(2x^2 - 3x - 2)}{2x^3 - 3x^2 - 2x}$$

(A1)

$$= 1 + \frac{2}{x}$$

(A1)

(3)

[7]

Q7.

(a) $f(-3) = (-3)^3 - 4x(-3) + 15$

f(-3) attempted not long division

M1

$$f(-3) = -27 + 12 + 15 = 0 \Rightarrow x + 3 \text{ is a factor}$$

shown = 0 plus statement

A1

2

(ii) Quadratic factor $(x^2 - 3x + 5)$

-3x or + 5 term by inspection or full long division attempt

$$(f(x) =) (x + 3)(x^2 - 3x + 5)$$

must see correct product

A1

2

(b) (i) $\left(\frac{dy}{dx} =\right) 4x^3 - 16x + 60$

one of these terms correct

M1

another term correct

A1

all correct (no +c etc)

A1

3

(ii) $4x^3 - 16x + 60 = 0$

must see this line OE

$$\Rightarrow x^3 - 4x + 15 = 0$$

AG

B1

1

- (iii) Discriminant of quadratic = $(-3)^2 - 4 \times 5$
discriminant of "their" quadratic or correct use of quad eqn "formula"

M1

$$\left. \begin{array}{l} b^2 - 4ac = -11 \text{ (or } b^2 - 4ac < 0 \text{)} \\ \text{therefore quadratic has no (real) roots} \\ \text{Hence only stationary point is when } x = -3 \end{array} \right\}$$

correct discriminant evaluated correctly
 (or shown to be < 0) with appropriate conclusion
plus final statement

A1

2

(iv) $\left(\frac{d^2y}{dx^2} = \right) 12x^2 - 16$

$= 12(-3)^2 - 16$ (or $12 \times 9 - 16$ etc)

sub $x = -3$ into "their" $\frac{d^2y}{dx^2}$

$= 92$

B1 ✓

M1

A1

3

- (v) Minimum **since** $\frac{d^2y}{dx^2} > 0$ (or $92 > 0$ etc)
FT appropriate conclusion from their value from (iv) plus reason
treat parts (iv) & (v) holistically

E1 ✓

1

[14]

Q8.

(a) $f\left(-\frac{3}{2}\right) = 4\left(-\frac{3}{2}\right)^3 - 11\left(-\frac{3}{2}\right) - 3$
 $x = -\frac{3}{2}$ substituted

M1

$$= -4 \times \frac{27}{8} + \frac{33}{2} - 3 = 0 \Rightarrow \text{factor}$$

Processing, = 0 and conclusion

A1

2

(b) $2x^2 - 3x - 1$

M1 for any two of a, b, c correct

M1A1

2

(c) (i) $2\cos 2\theta \sin \theta + 9\sin \theta + 3$
 $= 2(1 - 2\sin^2 \theta)\sin \theta + 9\sin \theta + 3$

cos 2θ expanded; ACF and substituted

M1

$$= 2\sin \theta - 4\sin^3 \theta + 9\sin \theta + 3$$

All in terms of sinθ or x and simplified to a cubic expression

m1

$$\sin \theta = x \Rightarrow 4x^3 - 11x - 3 = 0$$

*Reverse signs and express in x correctly **AG***

A1

3

(ii) $2x^2 - 3x - 1 = 0 \Rightarrow x = \frac{3 \pm \sqrt{17}}{4}$

Use formula correctly to solve $ax^2 + bx + c = 0$ from part (b)

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M1

$$x = \frac{3 - \sqrt{17}}{4} \text{ or } -0.28...$$

A1

$$\theta = 196^\circ \text{ and } 344^\circ$$

Both required and no others in range; condone greater accuracy. Ignore solutions out of range.

A1

$$x = \frac{3 + \sqrt{17}}{4} \text{ no solutions for } \sin \theta$$

$x = -\frac{3}{2}$ no solutions for $\sin \theta$
 Must have three correct roots and reject both other roots
 from cubic equation

E1

4

[11]

Q9.

(a) $16\left(\frac{3}{4}\right)^3 + 11\left(\frac{3}{4}\right) - 15$

Evaluate $f\left(\frac{3}{4}\right)$ not long division.

M1

$$= \frac{27}{4} + \frac{33}{4} - 15 = 0 \Rightarrow \text{factor}$$

Processing and conclusion.

A1

For A1; minimum processing seen; $16 \times \frac{27}{64} + 11 \times \frac{3}{4} - 15 = 0$; $15 - 15 = 0$

and no other working is A0 minimum conclusion = 0 hence factor

2

(b) $27\cos\theta(2\cos^2\theta - 1) +$

Use acf of $\cos 2\theta$ formula

B1

$$19\sin\theta(2\sin\theta\cos\theta) - 15 = 0$$

Use acf of $\sin 2\theta$ formula

B1

$$54\cos^3\theta - 27\cos\theta + 38(1 - \cos^2\theta)\cos\theta - 15 = 0$$

All in cosines.

M1

$$16\cos^3\theta + 11\cos\theta - 15 = 0$$

$$x = \cos\theta \Rightarrow 16x^3 + 11x - 15 = 0$$

Simplification and substitute $x = \cos\theta$ to obtain AG CSO.

A1

For M1 mark; $\cos 2\theta$ (eventually) in form $a \cos^2 \theta + b$; $19 \sin \theta \cos \theta$ in form $c \cos \theta \sin^2 \theta$ and use $\sin^2 \theta = 1 - \cos^2 \theta$ to obtain $c \cos \theta (1 - \cos^2 \theta)$

4

(c) $16x^3 + 11x - 15 = (4x - 3)(4x^2 + 3x + 5)$

Factorise $f(x)$

M1A1

$$b^2 - 4ac = 3^2 - 4 \times 4 \times 5 \quad (= -71)$$

Find discriminant of quadratic factor; or seen in formula

m1

$$b^2 - 4ac \quad x^2 + 3x + 5 = 0)$$

Conclusion; CSO

$$\Rightarrow \text{(only) solution is } \cos \theta = \frac{3}{4}$$

Condone $\frac{3}{4}$ is (only) solution

A1

M1 $(4x - 3)(4x^2 + kx \pm 5)$ A1 fully correct

m1 candidate's values of a , b , c used in expression for $b^2 - 4ac$

or complete square to obtain $\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$

A1 $b^2 - 4ac$ correct or $\left(x + \frac{3}{8}\right)^2 = \frac{9}{64} - \frac{5}{4} \quad \left(= -\frac{71}{64}\right)$ and stated

to be negative so no solution or solutions are not real (imaginary)
Accept imaginary solutions from calculator if stated to be imaginary.

Condone $\sqrt{-71}$ is negative, or similar, so no solution.

Conclusion $x = \frac{3}{4}$ is solution, or $\cos \theta = \frac{3}{4}$ is solution.

4

[10]

Q10.

(a) (i) $p(-1) = (-1)^3 + 2(-1)^2 - 5(-1) - 6$
 $p(-1)$ attempted **not** long division

M1

$$p(-1) = -1 + 2 + 5 - 6 = 0 \Rightarrow x + 1 \text{ is a factor}$$

CSO; correctly shown = 0 plus statement

A1

2

- (ii) Quad factor in this form: $(x^2 + bx + c)$
long division as far as constant term
or comparing coefficients,
or $b = 1$ or $c = -6$ by inspection

M1

$$x^2 + x - 6$$

correct quadratic factor

A1

$$[p(x) =] (x + 1)(x + 3)(x - 2)$$

must see correct product

A1

3

- (b) $p(0) = -6$; $p(1) = -8$
both $p(0)$ and $p(1)$ attempted
and at least one value correct

M1

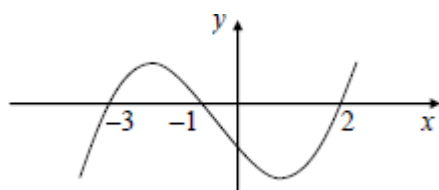
$$\Rightarrow p(0) > p(1)$$

AG both values correct plus correct statement
 involving $p(0)$ and $p(1)$

A1

2

(c)



cubic with one max and one min

M1

with $-3, -1, 2$ marked

A1

*correct with minimum to right of
 y-axis AND going beyond -3 and 2*

A1

3

[10]

Q11.

(a) (i) $5x - 6 = A(x - 3) + Bx$

Multiply by denominator and use two values of x .

M1

$$\begin{aligned} x = 0 & \quad x = 3 \\ A = 2 & \quad B = 3 \end{aligned}$$

A1

Alternative: equate coefficients

$$-6 = -3A \quad 5 = A + B$$

Set up and solve simultaneous equations for values of A and B .

(M1)

$$A = 2 \quad B = 3$$

(A1)

2

(ii) $\left(\int \frac{2}{x} + \frac{3}{x-3} dx \right) = 2 \ln x$
their $A \ln x$

B1ft

$$+ 3 \ln(x-3) (+C)$$

their $B \ln(x-3)$ and no other terms; condone $B \ln x - 3$

B1ft

2

(b) (i)

$$\begin{array}{r} 2x^2 - x + 3 \\ 2x+1 \overline{) 4x^3 + 5x - 2} \\ \underline{4x^3 + 2x^2} \\ -2x^2 + 5x - 2 \\ \underline{-2x^2 - x} \\ 6x - 2 \\ \underline{6x + 3} \\ -5 \end{array}$$

Division as far as $2x^2 + px + q$ with $p \neq 0$, $q \neq 0$, PI

M1

$$p = -1$$

PI by $2x^2 - x + q$ seen

A1

$$q = 3$$

PI by $2x^2 - x + 3$ seen

A1

$$r = -5$$

and must state $p = -1$, $q = 3$, $r = -5$ explicitly or write out full correct RHS expression

A1

Alternative 1:

$$4x^3 + 5x - 2 = 4x^3 + (2 + 2p)x^2 + (p + 2q)x + q + r$$

$$2 + 2p = 0$$

Clear attempt to equate coefficients, PI by $p = -1$

(M1)

$$p + 2q = 5$$

$$q + r = -2$$

$$p = -1$$

(A1)

$$q = 3 \quad r = -5$$

(A1A1)

Alternative 2:

$$4x^3 + 5x - 2 = (2x + 1)(2x^2 + px + q) + r$$

$$x = -\frac{1}{2} \quad 4 \times \left(-\frac{1}{2}\right)^3 + 5\left(-\frac{1}{2}\right) + 2 = r$$

$$x = -\frac{1}{2} \text{ used to find a value for } r$$

(M1)

$$r = -5$$

(A1)

$$p = -1, q = 3$$

(A1A1)

(ii) $\left(\frac{4x^3 + 5x - 2}{2x + 1} = \right) 2x^2 + px + q + \frac{r}{2x + 1}$

M1

$$\frac{2}{3}x^3 - \frac{1}{2}x^2 + 3x + k \ln(2x + 1) (+ C)$$

ft on p and q

A1ft

$$\frac{2}{3}x^3 - \frac{1}{2}x^2 + 3x - \frac{5}{2} \ln(2x + 1) (+ C)$$

CSO

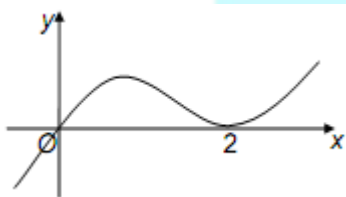
A1

3

[11]

Q12.

(a) (i)



*cubic curve with one max and one min
(either way up)*

M1

curve touching positive x-axis (either way up)

A1

*correct graph passing through O and touching
x-axis at 2*

A1

3

(ii) $x(x^2 - 4x + 4) = 3$

$$\Rightarrow x^3 - 4x^2 + 4x - 3 = 0$$

AG (must have = 0)

B1

1

(b) (i) $p(3) = 3^3 - 4 \times 3^2 + 4 \times 3 - 3$

p(3) attempted (condone one slip)

NOT long division

M1

$$p(3) = 27 - 36 + 12 - 3$$

$$p(3) = 0 \Rightarrow -3 \text{ is factor}$$

*shown = 0 **plus statement***

A1

2

(ii) Either $b = -1$ (coefficient of x correct)

or $c = 1$ (constant term correct)

*allow M1 for full attempt at long division or
comparing coefficients if neither b nor c is correct*

M1

$$p(x) = (x - 3)(x^2 - x + 1)$$

A1

2

(c) Discriminant of 'their quadratic' = $(-1)^2 - 4$

numerical expression must be seen

M1

Discriminant = -3 (or < 0) $\Rightarrow$ no real roots

*must have correct quadratic and statement
and all working correct*

A1cso

(Only real root is $x = 3$)

B1

3

[11]

Q13.

$$\begin{aligned}
 \text{(a) (i)} \quad f\left(-\frac{1}{3}\right) &= 9\left(-\frac{1}{3}\right)^3 + 18\left(-\frac{1}{3}\right)^2 - \left(-\frac{1}{3}\right) - 2 \\
 &= 9\left(-\frac{1}{27}\right) + 18\left(\frac{1}{9}\right) - \left(-\frac{1}{3}\right) - 2 \\
 &= -\frac{1}{3} + 2 + \frac{1}{3} - 2 = 0
 \end{aligned}$$

$$f\left(-\frac{1}{3}\right) \text{ attempted}$$

NOT long division

M1

$$\Rightarrow (3x + 1) \text{ is a factor}$$

Shown = 0 plus statement

A1

2

(ii) $(f(x) = (3x + 1)(3x^2 + kx - 2))$

3 and -2

M1

$$k = 5$$

A1

$$(f(x) = (3x + 1)(3x - 1)(x + 2))$$

A1

3

Alternative

Using long division

$$\begin{array}{r}
 3x^2 + 5x - 2 \\
 (3x + 1) \overline{) 9x^3 + 18x^2 - x - 2} \\
 \underline{9x^3 + 3x^2} \\
 15x^2 - x \\
 \underline{15x^2 + 5x} \\
 -6x - 2 \\
 \underline{-6x - 2} \\
 0
 \end{array}$$

$$3x^2 + ax + b$$

(M1)

$$3x^2 + 5x - 2$$

(A1)

$$(f(x) = (3x + 1)(3x - 1)(x + 2))$$

(A1)

(3)

(iii) $9x^3 + 21x^2 + 6x = x(9x^2 + 21x + 6)$

x and attempt to factorise quadratic equation.

M1

$$= 3x(3x+1)(x+2)$$

Correct factors

A1

$$\frac{9x^3 + 21x^2 + 6x}{f(x)} = \frac{3x}{3x-1}$$

cs0 no ISW

A1

3

Alternative

$$\frac{f(x) + q(x)}{f(x)}, \text{ where } q \text{ is a quadratic expression}$$

(M1)

$$= 1 + \frac{(3x+1)(x+2)}{(3x+1)(3x-1)(x+2)}$$

(A1)

$$= 1 + \frac{1}{3x-1}$$

(A1)

(3)

(b) $9\left(\frac{2}{3}\right)^3 + p\left(\frac{2}{3}\right)^2 - \frac{2}{3} - 2 = -4$

Condone missing brackets, but must have $= -4$

M1

$$p = -9$$

A1

(2)

[10]

Q14.

(a) $p(-1) = (-1)^3 - 2(-1)^2 + 3$

$p(-1)$ attempted; not long division

M1

$$p(-1) = -1 - 2 + 3 = 0 \Rightarrow x + 1 \text{ is a factor}$$

correctly shown = 0 plus statement

A1cso

2

- (b) (i) Quadratic factor ($x^2 - 3x + 3$)

b = -3 or c = 3 by inspection

or full long division attempt

or comparing coefficients

M1

$$(p(x) =) (x + 1)(x^2 - 3x + 3)$$

must see correct product

A1

2

- (ii) Discriminant of quadratic

$$b^2 - 4ac = (-3)^2 - 4 \times 3$$

'their' discriminant considered possibly within quadratic equation formula

M1

$$\left. \begin{aligned} b^2 - 4ac < 0 &\Rightarrow \text{no real roots from quadratic} \\ \Rightarrow \text{only one real root} \end{aligned} \right\}$$

A1cso

2

[6]

Q15.

- (a) $(f(-2) =)0$

ISW (0 seen is B1)

B1

1

(b)
$$f\left(\frac{3}{2}\right) = 4\left(\frac{3}{2}\right)^3 - 13\left(\frac{3}{2}\right) + 6$$

*Clear attempt at $f\left(\frac{3}{2}\right)$ with 3 terms
Factor theorem required;*

NOT long division

M1

$$4 \times \frac{27}{8} - 13 \times \frac{3}{2} + 6 \text{ or } 13.5 - 19.5 + 6$$

$$= 0 \Rightarrow (2x - 3) \text{ is a factor}$$

Must see this, or equivalent
Shown = 0 and statement.

A1

2

- (c) Any appropriate method to find third factor

Full long division

Compare coefficients

Factor Theorem $f\left(\frac{1}{2}\right)$

M1

$$(x + 2)(2x - 3)(2x - 1)$$

$$\text{Or } (2x^2 + x - 6)(2x - 1)$$

NMS M1A1

SC1 $(2x + 1)$ or $(1 - 2x)$ or $\left(x - \frac{1}{2}\right)$ or $\left(\frac{1}{2} - x\right)$
for third factor

A1

$$2x^2 + x - 6 = (x + 2)(2x - 3)$$

*Factorise **numerator** correctly or cancel $2x^2 + x - 6$*

M1

$$\frac{2x^2 + x - 6}{f(x)} = \frac{1}{2x - 1}$$

No ISW

A1

4

[7]

Q16.

(a) $(5 + h)^3 = 125 + 75h + 15h^2 + h^3$

Accept unsimplified coefficients

B1

1

- (b) (i) $y(5 + h) = 100 + 65h + 14h^2 + h^3$
PI; ft numerical error in (a)

B1F

Use of correct formula for gradient

M1

Gradient is $65 + 14h + h^2$

A1 if one numerical error made;

ft numerical error already penalised

A2,1F

4

- (ii) As $h \rightarrow 0$ this $\rightarrow 65$
E1 for 'h = 0';
ft wrong values for p, q, r

E2,1F

2

[7]

Q17.

- (a) $p - 3 = (-3)^3 - 13(-3) - 12$
must attempt $p - 3$ NOT long division

M1

$$= -27 + 39 - 12$$

$$= 0 \Rightarrow x + 3 \text{ is factor}$$

shown = 0 plus statement

A1

2

- (b) $x + 3 \quad x^2 + bx + c$
Full long division, comparing coefficients
or by inspection either $b = -3$ or $c = -4$

M1

$$x^2 - 3x - 4 \text{ obtained}$$

or M1 A1 for either $x - 4$ or $x + 1$

clearly found using factor theorem

A1

$$x + 3 \quad x - 4 \quad x + 1$$

CSO; must be seen as a product of 3 factors

NMS full marks for correct product

SC B1 for $x + 3 \quad x - 4$

or $(x + 3)(x + 1)(\quad)$

or $(x + 3)(x + 4)(x - 1)$ NMS

A1

3

[9]

Q18.

(a) (i) $f(-1) = -15 + 19 - 4 = 0$

B1

1

(ii) $f\left(\frac{2}{5}\right)$

evaluate **or** complete division leading to a numerical remainder

M1

$$\left(15 \times \frac{8}{125} + 19 \times \frac{4}{25} - 4\right) = 0 \Rightarrow \text{factor}$$

Or decimal equivalent $(0.96 + 3.04 - 4)$

or zero remainder $\Rightarrow$ factor

A1

2

(b) $(x + 1)$ is a factor

Stated or implied.

B1

Third factor is $(3x + 2)$

Any appropriate method to find third factor

M1A1

$$\frac{15x^2 - 6x}{f(x)} = \frac{3x(5x - 2)}{(x + 1)(5x - 2)(3x + 2)}$$

$(5x - 2)(3x^2 \pm 5x \pm 2)$ + attempt to factorise

Factorise numerator correctly

and attempt to simplify

M1

$$= \frac{3x}{(x+1)(3x+2)}$$

CSO no ISW

A1

5

[8]

Q19.

(a) $10x^2 + 8 = 2(x+1)(5x-1) +$

A and B terms correct

M1

$$A(5x-1) + B(x+1)$$

A1

$$x = -1 \quad x = \frac{1}{5}$$

$$A = -3 \quad B = 7$$

*Use two values of x to find A and B, or
set up and solve*

$$8 + 5A + B = 0$$

$$-2 - A + B = 8$$

SC1 NWS A & B correct $\frac{4}{4}$

SC2 NWS A or B correct $\frac{1}{4}$

m1A1

4

(b) $\int \frac{10x^2 + 8}{(x+1)(5x-1)} dx = \int 2 - \frac{3}{x+1} + \frac{7}{5x-1} dx$

Use the partial fractions

M1

$$= 2x + C$$

B1

*a ln (x + 1) + b ln (5x - 1)
condone missing brackets*

M1

$$-3 \ln(x+1) + \frac{7}{5} \ln(5x-1)$$

F on A and B

A1F

4

[8]

Q20.

(a) (i) $p(-3) = (-3)^3 + 7(-3)^2 + 7(-3) - 15$
p(-3) attempted; NOT long division

$$= -27 + 63 - 21 - 15$$

This line alone implies M1

M1

$$p(-3) = 0 \Rightarrow (x+3 \text{ is) factor}$$

p(-3) shown = 0 plus statement

A1

2

(ii) $p(x) = (x+3)(x^2 + px + q)$
*Full long division, comparing coefficients
 or by inspection either $p = 4$ or $q = -5$*

M1

(Quadratic factor) $(x^2 + 4x - 5)$
*or M1A1 for either $x - 1$ or $x + 5$
 clearly found using Factor Theorem*

A1

$(p(x) =) (x+3)(x-1)(x+5)$
*Must be seen as a product of 3 factors
 NMS full marks for correct product
 SC B2 for 3 correct factors listed NMS
 SC B1 for $(x+3)(x-1)()$
 or $(x+3)(x+5)()$
 or $(x+3)(x+1)(x-5)$*

A1

3

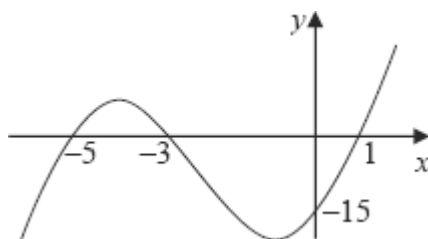
(b) (i) $p(-1) = -16; p(0) = -15$
 $\Rightarrow p(-1) < p(0)$

Values must be evaluated correctly

B1

1

(ii)



y-intercept -15 marked or (0, -15) stated

B1

Cubic graph – 1 max, 1 min

M1

~ shape with -5, -3, 1 marked

A1

*Graph correct with minimum point to left of y-axis and going beyond both -5 and 1
Previous A1 must be scored*

A1

Cannot score M1A0A1 but can score B0M1A1A1

4

[10]

Q21.

(a) $g\left(\frac{1}{4}\right) = \text{number}(s) + d = 0$

*Use factor theorem to find d
See some processing*

M1

$d = 3$

NMS 2/2

A1

Alternative:

Division $\Rightarrow d - 3$ last line
Candidate's - 3

(M1)

$$d = 3$$

(A1)

2

(b) $g(x) = (4x - 1)(2x^2 + bx - 3)$
 $a = 2 \quad c = -3; \quad F \text{ on } d \ (c = -d)$

B1F

$$x^2 - 6 = 4b - 2 \quad \text{or} \quad x^2 - 14 = -b - 12$$

Any appropriate method; PI

M1

$$b = 2$$

NMS 2/2

A1

3

[5]

Q22.

(a) (i) $7x - 3 = A(3x - 2) + B(x + 1)$

M1

$$x = -1 \quad x = \frac{2}{3}$$

Substitute two values of x and solve for A and B

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m1

$$A = 2 \quad B = 1$$

Or solve
$$\begin{cases} 7 = 3A + B \\ -3 = -2A + B \end{cases} \quad \text{condone one error}$$

A1

Alternative:

By cover up rule

$$x = -1 \quad A = \frac{-7 - 3}{-5}$$

$$x = \frac{2}{3} \quad B = \frac{\frac{14}{3} - 3}{\frac{5}{3}}$$

$$x = -1 \quad \text{and} \quad x = \frac{2}{3}$$

and attempt to find A and B

(M1)

$$A = 2 \quad B = 1$$

SC NMS A and B both correct 3/3

One of A or B correct 1/3

(A1, A1)

3

(ii) $\int \frac{7x-3}{(x+1)(3x-2)} dx =$

$$p \ln(x+1) + q \ln(3x-2)$$

Condone missing brackets

M1

$$= 2 \ln(x+1) + \frac{1}{3} \ln(3x-2) (+c)$$

F on A and B ; constant not required

A1F

2

(b) $\frac{6x^2 + x + 2}{2x^2 - x + 1} = \frac{6x^2 - 3x + 3 + 4x - 1}{2x^2 - x + 1}$

$$= 3 + \frac{4x-1}{2x^2-x+1}$$

$$P = 3$$

M1

B1

$$Q = 4 \text{ and } R = -1$$

A1

Alternative:

$$\begin{array}{r}
 3 \\
 2x^2 - x + 1 \overline{) 6x^2 + x + 2} \\
 \underline{6x^2 - 3x + 3} \\
 4x - 1
 \end{array}$$

Complete division, with $ax + b$ remainder

(M1)

$$P = 3 \text{ stated}$$

(B1)

$Q = 4$ and $R = -1$ stated or written as expression

(A1)

or

$$6x^2 + x + 2 = P(2x^2 - x + 1) + Qx + R$$

$$= 2Px^2 + (Q - P)x + P + R$$

Multiply across and equate coefficients or use numerical values of x

(M1)

$$P = 3$$

$P = 3$ stated

(B1)

$$Q - P = 1$$

$$P + R = 2$$

$$Q = 4 \text{ and } R = -1$$

$Q = 4$ and $R = -1$ stated or written as expression

(A1)

3

[8]

Q23.

(a) (i) $p(2) = 8 + 2 - 10$

Must find $p(2)$ NOT long division

M1

$$\Rightarrow p(2) = 0 \Rightarrow (x - 2) \text{ is factor}$$

Shown = 0 plus a statement

A1

2

(ii) Attempt at long division (generous)

Obtaining a quotient $x^2 + cx + d$
or equating coefficients (full method)

M1

$$p(x) = (x - 2)(x^2 + 2x + 5)$$

$a = 2$, $b = 5$ by inspection B1, B1

A1

(b) (i) $\frac{dy}{dx} = 3x^2 + 1$

One term correct

M1

All correct – no + c etc

A1

When $x = 2 \frac{dy}{dx} = 3 \times 4 + 1$

Sub $x = 2$ into their $\frac{dy}{dx}$

m1

Therefore gradient at Q is 13
CSO

A1

4

(ii) $y = 13(x - 2)$

*Tangent (NOT normal) attempted
ft their gradient answer from (b)(i)*

M1

CSO; correct in any form

A1

2

(iii) $\int \dots dx = \frac{x^4}{4} + \frac{x^2}{2} - 10x (+c)$

one term correct

M1

second term correct

A1

all correct (condone no +c)

A1

3

(iv) $[4 + 2 - 20] - [0] = -14$

*F(2) attempted and possibly F(0)
Must have earned M1 in (b)(iii)*

M1

Area of shaded region = 14

CSO; separate statement following correct evaluation of limits

A1

2

[15]

Q24.

(a) $f(-1) = 0$

B1

1

(b) $f\left(-\frac{1}{2}\right) = 4\left(-\frac{1}{8}\right) - 7\left(-\frac{1}{2}\right) - 3$

Use of $\pm \frac{1}{2}$

M1

$= -\frac{1}{2} + \frac{7}{2} - 3 = 0 \Rightarrow \text{factor}$

Need to see simplification (at least)

$\left(-\frac{1}{2}\right)^3 = -\frac{1}{8}, ' = 0' \text{ and conclusion}$

A1

2

(c) Third factor is $(2x - 3)$

PI

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B1

$$\frac{(x+1)(2x+1)(2x-3)}{(x+1)(2x+1)}$$

3 linear factors
2 linear factors

M1

simplifies to $2x - 3$

Simplified result stated.

Alternative; see end.

Use remainder theorem.

A1

Alternative

Complete division to $2x + b$

(M1)

Complete division to $2x - 3$

(A1)

simplifies to $2x - 3$

Simplified result stated

(A1)

3

[6]

Q25.

(a) (i) $\frac{2x+7}{x+2} = 2 + \frac{3}{x+2}$

B1B1

2

(ii) $\int \frac{2x+7}{x+2} = 3 \ln(x+2) + 2x + C$

Either term correct

B1F

*Both correct; constant required; condone missing bracket
ft on A, B*

B1F

2

(b) (i) $28 + 4x^2$
 $P(5-x)^2 + Q(1+3x)(5-x) + R(1+3x)$

M1

$x = 5$ $x = -\frac{1}{3}$

Two values of x used to find R and P.

m1

$R = 8$ $P = 1$

SC R = 8, P = 1 NMS can score B1, B1

A1

$$x = 0 \Rightarrow 28 = 25P + 5Q + R$$

Third value of x used to find Q

m1

$$Q = -1$$

A1

Alternative

$$28 + 4x^2 =$$

$$P(5-x)^2 + Q(1-3x)(5-x) + R(1+3x)$$

(M1)

$$= (25P + 5Q + R) + (-10P + 14Q + 3R)x + (P - 3Q)x^2$$

Collect terms and form equations

(m1)

$$P - 3Q = 4$$

$$14Q + 3R - 10P = 0$$

Correct equations

(A1)

$$25P + 5Q + R = 28$$

$$P = 1 \quad Q = -1 \quad R = 8$$

Solve for P Q and R

(m1)(A1)

5

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(ii) $\int \frac{1}{1+3x} - \frac{1}{5-x} + \frac{8}{(5-x)^2} dx$

Use partial fractions

M1

$$= \frac{1}{3} \ln(1+3x) + \ln(5-x) + \frac{8}{5-x} + (C)$$

$$a \ln(1+3x) + b \ln(5-x)$$

m1

OE; both ln integrals correct; needs ()

A1F

Other term correct

ft on their P, Q, R

A1F

4

SC: If no P, Q, R found in (b)(i), can gain method marks by inserting other values or retaining the letters (max 2/4)

[13]

Q26.

(a) (i) $p(-2) = -8 + 2 + 6$

$p(-2)$ attempted : **NOT** long division

M1

$p(-2) = 0 \Rightarrow x + 2$ is factor

Shown = 0 plus statement

May make statement first such as " $x + 2$ is a factor if $p(-2) = 0$ "

A1

Minimum statement required "factor"

2

(ii) $b = -2$

No working required for B1 + B1

B1

$c = 3$

Try to mark first using B marks

B1

or long division/comparing coefficients

Award M1 if B0 earned and a clear method is used

(M1)

$p(x) = (x + 2)(x^2 - 2x + 3)$

Must write final answer in this form if long division has been used to get A1

(A1)

2

(iii) $b^2 - 4ac = (-2)^2 - 4 \times 3$

Discriminant correct from their quadratic

M0 if $b = -1, c = 6$ used (using cubic eqn)

M1

$b^2 - 4ac = -8$ (or < 0) $\Rightarrow$ no (other) real roots
CSO All values must be correct plus statement

A1

Or $(x - 1)^2 + 2$
Completion of square for their quadratic

(M1)

$(x - 1)^2 + 2 > 0$ therefore no real roots
Shown to be positive plus statement regarding no real roots

(A1)

Or $(x - 1)^2 = -2$ has no real roots

2

(b) (i) $(y_B =) 6$
Condone (0, 6)

B1

1

(ii) $\frac{x^4}{4} - \frac{x^2}{2} + 6x$
One term correct
Another term correct

M1

A1

All correct (ignore + c or limits)

A1

$$\left[\begin{array}{c} 0 \\ -2 \end{array} \right] = 0 - (4 - 2 - 12)$$

$$= 10$$
F(-2) attempted

m1

$= 10$
CSO Clearly from $F(0) - F(-2)$

A1

5

(iii) Area of $\Delta = \frac{1}{2} \times 2 \times 6$

Condone – 2 and ft their y_B value

Or $\int_{-2}^0 (3x + 6) dx$ and attempt to integrate

M1

$= 6$

Must be positive allow –6 converted to +6

A1

Shaded region area = $10 - 6 = 4$

CSO 10 must come from correct working

A1

3

[15]

Q27.

$$\begin{array}{r}
 \overline{) 3x^3 + 8x^2 - 3x - 5} \\
 \underline{3x^3 - x^2} \\
 9x^2 - 3x \\
 \underline{ 9x^2 - 3x} \\
 0 - 5
 \end{array}$$

Division with x^2 and an x term seen; $x^2 + px$

M1

$a = 1 \quad b = 3 \quad \text{or} \quad x^2 + 3x + \frac{c}{3x-1}$

Explicit or in expression

A1

$c = -5$

Condone $+\frac{-5}{3x-1}$

B1

Alternative

$$\frac{(3x-1)(x^2+px)}{3x-1} - \frac{5}{3x-1}$$

Split fraction and attempt factors

(M1)

$$x^2 + 3x - \frac{5}{3x-1}$$

$$a = 1 \quad b = 3$$

(A1)

$$c = -5$$

(B1)

Alternative

$$f(x) = 3ax^3 + (3b - a)x^2 - bx + c$$

Multiply by $(3x - 1)$ and attempt to collect terms

(M1)

$$a = 1 \quad b = 3$$

(A1)

$$c = -5$$

(B1)

Alternative

$$f(x) = (ax^2 + bx)(3x - 1) + c$$

Multiply by $(3x - 1)$ and attempt to find a, b, c : substitute 3 values of x and form 3 simultaneous equations, and attempt to solve; or substitute 3 values of x into given equation

(M1)

$$x = 0 \Rightarrow c = -5$$

(B1)

$$x = 1 \Rightarrow 2a + 2b + c = 3$$

$$x = 2 \Rightarrow 20a + 10b + c = 45$$

$$a = 1 \quad b = 3$$

(A1)

[3]

Q28.

(a) (i) $p(-1) = -1 + 7 - 6$
Finding $p(-1)$

M1

$= 0$ therefore $x + 1$ is a factor
*Shown to = 0 **plus statement***

A1

2

(ii) $p(x) = (x + 1)(x^2 - x - 6)$
Long division/inspection (2 terms correct)

M1

Quadratic factor correct

A1

$p(x) = (x + 1)(x + 2)(x - 3)$
*May earn M1, A1 For correct second factor
 then A1 For $(x + 1)(x + 2)(x - 3)$*

A1

3

(b) (i) $A(-2, 0)$
Condone $x = -2$

B1

1

(ii) $\frac{x^4}{4} - \frac{7x^2}{2} - 6x (+c)$
One term correct

M1

Another term correct

A1

(may have $+ c$ or not)
All correct unsimplified

A1

$\left[\frac{81}{4} - \frac{63}{2} - 18 \right] - \left[\frac{1}{4} - \frac{7}{2} + 6 \right]$

$F(3) - F(-1)$ attempted in correct order

m1

$$= -32$$

CSO; OE

A1

5

(iii) Area of shaded region = 32

FT their (b)(ii) but positive value needed

B1ft

1

(iv) $\frac{dy}{dx} = 3x^2 - 7$

One term correct

M1

All correct (no + c etc)

A1

When $x = -1$, gradient = -4

CSO

A1

3

(v) Gradient of normal = $\frac{1}{4}$

B1ft

$y = \text{"their gradient"} (x \pm 1)$

*Must be finding **normal**, not tangent*

M1

$$y = \frac{1}{4}(x + 1)$$

CSO; any correct form eg $4y - x = 1$

A1

3

[18]

Q29.

(a) (i) $f\left(\frac{1}{2}\right) = 2 \times \left(\frac{1}{2}\right)^3 + 3 \times \left(\frac{1}{2}\right)^2 - 18\left(\frac{1}{2}\right) + 8$

use of $\pm \frac{1}{2}$ substituted in $f(x)$

M1

$$= \frac{1}{4} + \frac{3}{4} - 9 + 8 = 0 \Rightarrow \text{factor}$$

arithmetic seen and conclusion –

minimum seen: $2 \times \frac{1}{8} + 3 \times \frac{1}{4} - 18 \times \frac{1}{2} + 8 = 0$

A1

2

(ii) $f(x) = (2x - 1)(x^2 + 2x - 8)$
or $p = 2, q = -8$

B1B1

2

(iii) $\frac{4x(x+4)}{(2x-1)(x+4)(x-2)}$
numerator correct; attempt to factorise denominator (algebraic fraction not required)

M1

$$= \frac{4x}{(2x-1)(x-2)}$$

CAO

A1

2

(b) $2x^2 = A(x+5)(x-3) + B + Cx$
any equivalent method
using PFs (see alternative method)

M1

$$A = 2$$

B1

$$2A + C = 0 \quad -15A + B = 0$$

equate coefficients or use 2 values of x to find B and C

M1

$$C = -4 \quad B = 30$$

both B and C correct

A1

ALTERNATIVE METHOD 1

$$\begin{array}{r}
 x^2 + 2x - 15 \overline{) 2x^2} \\
 \underline{2x^2 + 4x - 30} \\
 -4x + 30
 \end{array}$$

complete division

(M1)

$$A = 2$$

(B1)

$$B = 30$$

(A1)

$$C = -4$$

(A1)

ALTERNATIVE METHOD 2

$$\begin{aligned}
 \frac{2x^2}{(x+5)(x-3)} &= A + \frac{D}{x+5} + \frac{E}{x-3} \\
 2x^2 &= A(x+5)(x-3) + D(x-3) + E(x+5)
 \end{aligned}$$

$$\begin{aligned}
 x = 3 \quad 18 &= 8E \quad E = \frac{9}{4}
 \end{aligned}$$

$$\begin{aligned}
 x = -5 \quad 50 &= -8D \quad D = -\frac{25}{4}
 \end{aligned}$$

find D and E

(M1)

$$x = 0, 0 = -15A + \left(-\frac{25}{4}\right)(-3) + \left(\frac{9}{4}\right)(5)$$

$$A = 2$$

(B1)

$$\begin{aligned}
 \frac{D}{x+5} + \frac{E}{x-3} &= \frac{-25}{4(x+5)} + \frac{9}{4(x-3)} \\
 &= \frac{-25(x-3) + 9(x+5)}{4(x+5)(x-3)} \\
 &= \frac{120 - 16x}{4(x+5)(x-3)}
 \end{aligned}$$

recombine to required form

(M1)

$$= \frac{30 - 4x}{(x+5)(x-3)}$$

CAO

(A1)

4

[10]

Q30.

(a) (i) $p(-2) = -8 + 4 + 16 - 12$

NOT long division

M1

$= 0 \Rightarrow (x + 2)$ is factor

p(-2) shown = 0 and statement

A1

2

(ii) Quad factor by comparing coefficients or
 $(x^2 + kx \pm 6)$ by inspection

Or full long division or attempt at Factor Theorem using f(±3)

M1

$p(x) = (x + 2)(x^2 - x - 6)$

*Correct quadratic factor or $(x - 3)$ shown to be factor by
 Factor Theorem*

A1

$p(x) = (x + 2)^2(x - 3)$ or $(x + 2)(x + 2)(x - 3)$

*CSO; SC: B1 For $(x + 2)(x^{***})(x - 3)$ by inspection or without
 working*

A1

3

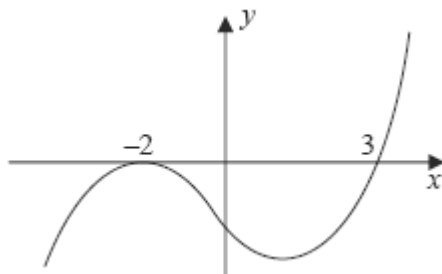
- (b) (i) $(k =) -12$

Condone $y = -12$ or $(0, -12)$

B1

1

- (ii)



Cubic shape (one max and one min)

M1

*Maximum at $(-2, 0)$ and through $(3, 0)$ –
at least one of these values marked*

A1

*“correct” graph as shown
(touching smoothly at -2 , 3 marked and
minimum to right of y -axis)*

A1

3

[9]

Q31.

(a) $f\left(-\frac{2}{3}\right) = -8 + 6 + 2 = 0$

AG

B1

1

(b) $f(x) = (3x + 2)(ax^2 + bx + c)$

$(3x + 2)$ or $\left(x + \frac{2}{3}\right)$ is a factor PI

B1

$a = 9 \quad c = 1$

*quadratic factor; find coefficients; 2 correct
equate coefficients and solve for b*

M1

$$x^2 \text{ term } 3b + 2a = 0$$

or

$$x \text{ term } 3c + 2b = -9$$

$$b = -6$$

$$\text{or (could be shown as) } 9x^2 - 6x + 1$$

correct quadratic factor or a , b , and c correct

or use division or factor theorem to seek another factor (see alternative methods at end of scheme)

A1

$$f(x) = (3x + 2)(3x - 1)(3x + 2)$$

SC (see alternative methods below)

A1

ALTERNATIVE METHOD 1

$(3x + 2)$ is a factor

PI

B1

use factor theorem

use factor theorem or algebraic division to find another factor

M1

$$f\left(\frac{1}{3}\right) = 0 \Rightarrow (3x - 1)$$

is a factor

$$f(x) = (3x + 2)(3x - 1)(ax + b)$$

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A1

$$f(x) = (3x + 2)(3x - 1)(3x - 1)$$

A1

ALTERNATIVE METHOD 2

$(3x + 2)$ is a factor

PI by division

B1

divide $27x^3 - 9x + 2$ by $(3x + 2)$

complete division to $ax^2 + bx + c$

M1

$$9x^2 - 6x + 1$$

A1

$$f(x) = (3x + 2)(3x - 1)(3x - 1)$$

A1

4

SPECIAL CASE

$$(3x + 2)(3x - 1)(ax + b)$$

2

(c) $9x^2 + 3x - 2 = (3x - 1)(3x + 2)$

*factorise denominator correctly or
complete division*

M1

$$\frac{27x^2 - 9x + 2}{9x^2 + 3x - 2} = 3x - 1$$

simplified result indicated

A1

2

[7]

Q32.

(a) (i) $p(-2) = -8 - 16 + 14 + k$
 $p(-2) = 0 \Rightarrow -10 + k = 0 \Rightarrow k = 10$

Must have statement if $k = 10$ substitute

or long division or $(x + 2)(x^2 - 6x + 5)$

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M1

AG likely withhold if $p(-2) = 0$ not seen

A1

2

(ii) $p(x) = (x + 2)(x^2 + \dots 5)$

$$p(x) = (x + 2)(x^2 - 6x + 5)$$

$$\Rightarrow p(x) = (x + 2)(x - 1)(x - 5)$$

Attempt at quadratic or second linear factor

M1

$(x - 1)$ or $(x - 5)$ from factor theorem

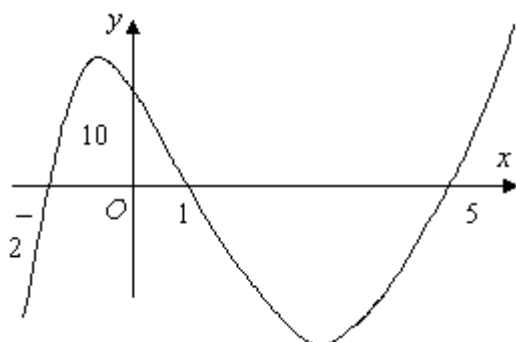
A1

Must be written as product

A1

3

(b)



Curve thro' 10 marked on y-axis

B1

FT their 3 roots marked on x-axis

B1 ✓

Cubic shape with a max and min

M1

Correct graph (roughly as on left) going beyond -2 and 5
(condone max anywhere between $x = -2$ and 1 and min between 1 and 5)

A1

4

[9]

Q33.

$$g\left(\frac{3}{2}\right) = 0 \Rightarrow d + 4 = 0 \Rightarrow d = -4$$

(a)

AG (convincingly obtained)

SC Written explanation with $g\left(\frac{3}{2}\right) = 0$
not seen/clear E2, 1, 0

M1A1

2

(b) $a = -2, \quad b = -3$

Inspection expected

By division: M1 – complete method

A1 CAO

Multiply out and compare coefficients:

M1 – evidence of use

A1 – both a and b correct

B1, B1

2

[4]

Q34.

(a) (i) $\frac{3x-5}{x-3} = 3 + \frac{4}{x-3}$

By division: B1 for 3, B1 for $\frac{4}{x-3}$ or $B = 4$
By partial fractions: M1 multiply by $x - 3$
and using 2 values of x , A1 both correct

B1, B1

2

(ii) $\int 3 + \frac{4}{x-3} dx = 3x + 4 \ln(x-3) (+c)$

M1 $\int 3 + \frac{4}{x-3} dx$ and attempt at integrals
ft on A and B; condone omission of
brackets around $x - 3$

M1A1F

2

Alternative: By substitution $u = x - 3$

$$\int \frac{3x-5}{x-3} dx = \int \frac{3u+4}{u} du$$

Integral in terms of u

(M1)

$$= 3(x-3) + 4 \ln(x-3)$$

Correct, in x

(A1)

(b) (i) $6x - 5 = P(2x - 5) + Q(2x + 5)$

Clear evidence of use of cover-up rule M2

M1

$$x = \frac{5}{2} \quad x = -\frac{5}{2}$$

m1

$$10 = 10Q - 20 = -10P$$

$$Q = 1 \quad P = 2$$

A1

3

(ii) $\int \frac{2}{2x+5} + \frac{1}{2x-5} dx$

Attempt at ln integral
(a ln (2x + 5) + b ln (2x - 5))

M1

$$\ln(2x+5) + \frac{1}{2} \ln(2x-5) (+c)$$

M1

ft on P and Q; must have brackets

A1F

3

[10]

Q35.

(a) (i) $f(1) = 1 + 4 - 5$

must find f(1) NOT long division

M1

$$\Rightarrow f(1) = 0 \Rightarrow (x - 1) \text{ is factor}$$

shown = 0 plus a statement

A1

2

(ii) Attempt at $x^2 + x + 5$

long division leading to $x^2 \pm x \dots$ or
equating coefficients

M1

$$f(x) = (x - 1)(x^2 + x + 5)$$

p = 1, q = 5 by inspection scores B1, B1

A1

2

(iii) $(x=)1$ is real root

B1

Consider $b^2 - 4ac$ for their $x^2 + x + 5$
not the cubic!

M1

$$b^2 - 4ac = 1^2 - 4 \times 5 = -19 < 0$$

Hence no real roots (or only real root is 1)
CSO; all values correct plus a statement

A1

3

(b) (i) $\int \dots dx = \frac{x^2}{4} + 2x^2 - 5x(+c)$

one term correct unsimplified

M1

second term correct unsimplified

A1

all correct unsimplified

A1

3

(ii) $[4+8-10] - \left[\frac{1}{4} + 2 - 5 \right]$

*correct use of limits 1 and 2;
 $F(2) - F(1)$ attempted*

M1

$$= 4 \frac{3}{4}$$

A1

Area of $\Delta = \frac{1}{2} \times 11 = 5 \frac{1}{2}$

correct unsimplified

B1

$$\Rightarrow \text{shaded area} = 5\frac{1}{2} - 4\frac{3}{4}$$

*combined integral of $7x - 6 - x^3$ scores
M1 for limits correctly used then*

$$= \frac{3}{4}$$

A3 correct answer with all working correct

A1

4

[14]



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